

## Section 04

# Industrial Security (Bachelor)



Section 04 – Industrial Protocols

## Industrial Protocols

*Lecture Notes*

Department of Computer Science

2026-05-18

Industrial Security (Bachelor)

Section 04

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Section 04 – Industrial Protocols

Industrial Protocols  
Lecture Notes  
Department of Computer Science

- 1 Legacy Heritage
- 2 Modbus
- 3 Siemens, DNP3, EtherNet/IP
- 4 Modern Secure Protocols

## By the end of this section you will be able to

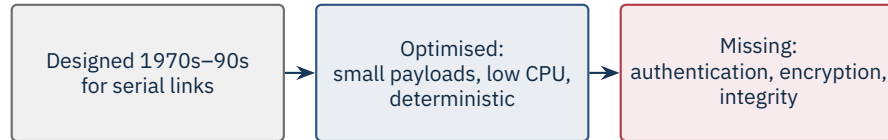
- Recognise common industrial protocols and their use cases.
- Understand why legacy protocols typically lack authentication.
- Decode a Modbus TCP frame by hand.
- Compare insecure legacy protocols with secure modern alternatives (OPC UA, MQTT-TLS).
- Identify when secure variants are appropriate.

# 1

## Legacy Heritage

## Core fact

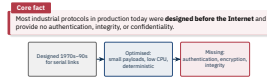
Most industrial protocols in production today were **designed before the Internet** and provide no authentication, integrity, or confidentiality.



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## Why Industrial Protocols Are Different

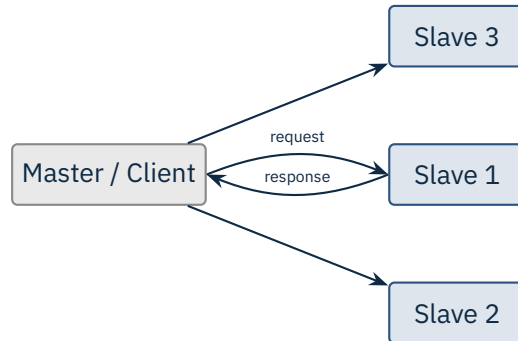
Set the framing before naming any single protocol: these wire formats predate routine internet use and were drawn for trusted serial loops where the physical layer *was* the security boundary. Push back on the student reflex that “newer = better” — the same Modbus frame from 1979 still runs in greenfield plants today because it is cheap, deterministic, and works. Misconception to flag early: “but it runs over TCP now, so it must have TLS somewhere.” No — TCP only gives you a byte stream; it does not authenticate anything. Transition cue: “so let us look at the canonical example, Modbus.”



# 2

## Modbus

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Variants: Modbus RTU (serial), Modbus TCP (port 502)  
Address space: coils, discrete inputs, input regs, holding regs  
No auth, no encryption – by design (1979)

*Source: Modbus Organization, Modbus Application Protocol Specification v1.1b3, 2012.*

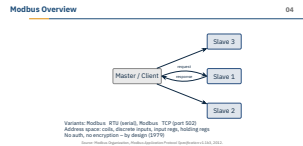
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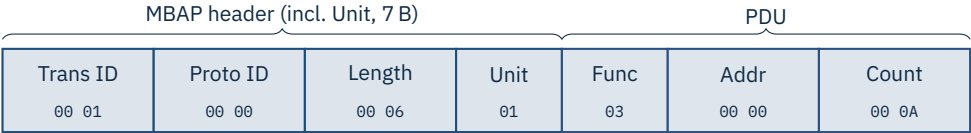
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└─ Modbus

└─ Modbus Overview

Anchor Modbus in time: it is older than ARP, older than DNS, older than most students in the room. The design assumption was a private 4-wire RS-485 loop on the shop floor, not the internet. When you see Modbus TCP today it is the original PDU with a 7-byte MBAP header bolted on — function code intact, no auth, no integrity. Ask the room: “Shodan shows tens of thousands of devices on port 502 — what would you check first?” Steer them toward unit-ID enumeration and function-code 8 (diagnostics) before anything destructive. Transition cue: “let us decode an actual frame so you can read it on the wire.”





Decoded

“Read 10 holding registers starting at address 0x0000 from slave 1.”

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└─Modbus

└─Modbus TCP Frame Anatomy

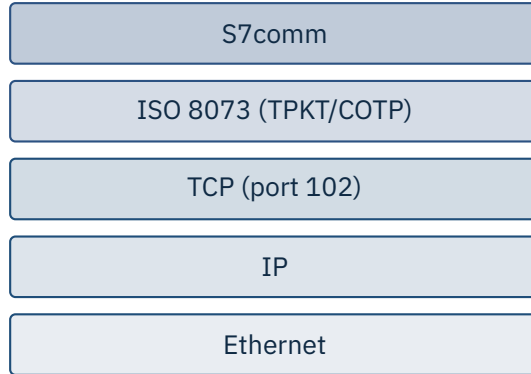
Walk the bytes left-to-right at the board: transaction ID is a client-chosen correlation cookie, protocol ID is always zero for Modbus, length counts everything from the unit byte onward. Stress that the unit ID was originally the RS-485 slave address and is now reused as a gateway selector — a single Modbus TCP endpoint can front a whole RTU bus. Common student misconception: they treat “Function 03” as read-only and therefore safe; remind them that Function 06 (single write) and 16 (multi-write) sit one byte away. Quick class question: “what changes in this frame if we want to write coils instead of read registers?” Answer: function code becomes 0x05 or 0x0F, plus a value field. Transition cue: “Modbus is the simple case — vendors like Siemens went their own way.”





# 3

## Siemens, DNP3, EtherNet/IP



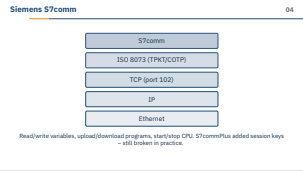
Read/write variables, upload/download programs, start/stop CPU. S7commPlus added session keys  
– still broken in practice.

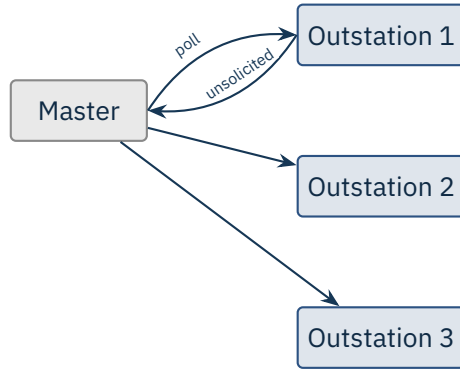
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└ Siemens, DNP3, EtherNet/IP

└ Siemens S7comm

Walk up the stack from Ethernet to S7comm so students see why classic packet captures show TPKT/COTP framing first — it is ISO transport over TCP because Siemens reused their pre-IP stack. The war story: S7comm has no authentication on classic S7-300/400, and the protocol was reverse-engineered well enough that Snap7 and python-snap7 give you full PLC control in a few lines. S7commPlus added an anti-replay handshake on S7-1200/1500, but researchers (Lei *et al.*, Biham, Plc-blaster) showed the key material was derivable; Siemens has rotated it several times. Misconception: “a firmware upgrade fixes it.” Often the upgrade silently changes the wire format and breaks third-party tools without actually adding cryptographic identity. Transition cue: “the utilities sector hit the same wall but answered differently.”





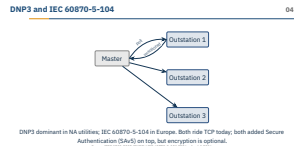
DNP3 dominant in NA utilities; IEC 60870-5-104 in Europe. Both ride TCP today; both added Secure Authentication (SAv5) on top, but encryption is optional.

Source: IEEE 1815-2012 (DNP3) | IEC 60870-5-104:2006 + Amd. 1:2016.

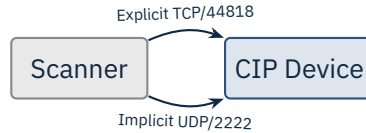
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Industrial Security (Bachelor)  
└ Siemens, DNP3, EtherNet/IP

└ DNP3 and IEC 60870-5-104



Highlight the master/outstation model and the *unsolicited response* pattern — outstations push event data back without being polled, which is essential when a substation has a fault to report. War story: DNP3 SAv5 was standardised in IEEE 1815-2012 with HMAC-based challenge/response, and over a decade later real-world adoption in the field is still measured in single-digit percentages; utilities cite key-management overhead and the fact that the link layer is often a leased serial circuit they already trust. Misconception: “SAv5 means encrypted.” No — it provides authentication only; confidentiality requires bump-in-the-wire TLS or IPsec. Class question: “why might a utility leave SAv5 off even after they buy capable RTUs?” Transition cue: “Rockwell took yet another approach with EtherNet/IP.”



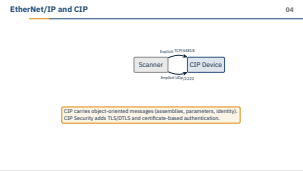
CIP carries object-oriented messages (assemblies, parameters, identity). CIP Security adds TLS/DTLS and certificate-based authentication.

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└ Siemens, DNP3, EtherNet/IP

└ EtherNet/IP and CIP

Make the split explicit: *explicit* messaging is request/response over TCP/44818 for configuration and diagnostics; *implicit* (a.k.a. I/O) is cyclic UDP/2222 multicast for the actual control data — which is why blocking 44818 alone does not stop a running line. CIP itself is an object model; EtherNet/IP is just CIP-over-Ethernet. War story: ODVA standardised CIP Security in 2015 with TLS for explicit and DTLS for implicit traffic, but Rockwell only ships it on newer ControlLogix 5580 / CompactLogix 5380 firmware, and rolling certificates across hundreds of drives is a project few sites have funded. Misconception: “EtherNet/IP is just industrial Ethernet.” No — the name is a marketing collision; it is CIP encapsulated in Ethernet frames, nothing to do with the EtherNet trademark. Transition cue: “Siemens did the same idea differently with Profinet.”

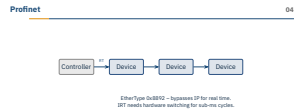




EtherType 0x8892 – bypasses IP for real time.  
IRT needs hardware switching for sub-ms cycles.

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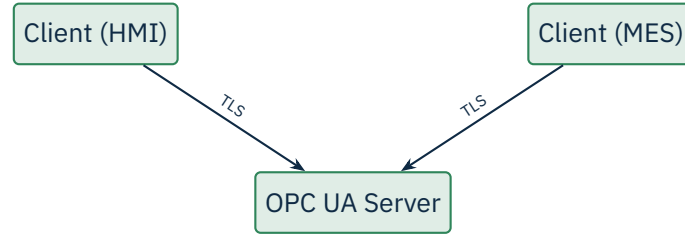
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└ Siemens, DNP3, EtherNet/IP  
  
└ Profinet



Stress that Profinet RT lives at layer 2 – EtherType 0x8892, no IP, no TCP – specifically because the IP stack adds jitter that breaks deterministic cycles. Profinet IRT goes further and requires switches with hardware time-slotting (typically IEEE 1588 PTP) for sub-millisecond updates. Security war story: Profinet has long-standing weaknesses around DCP (Discovery and Configuration Protocol), which lets anyone on the local segment re-name or reset a device; multiple advisories (CERT@VDE, Claroty’s OT:ICEFALL set in 2022) catalogued this. Misconception: “it is layer 2 so it cannot be attacked remotely.” True – but most Profinet networks share VLANs with the engineering laptops, and a compromised laptop is on the wire. Transition cue: “so what does a protocol look like when security is designed in from day one?”

# 4

## Modern Secure Protocols

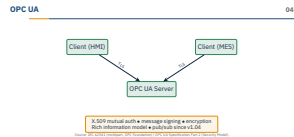


X.509 mutual auth • message signing • encryption  
Rich information model • pub/sub since v1.04

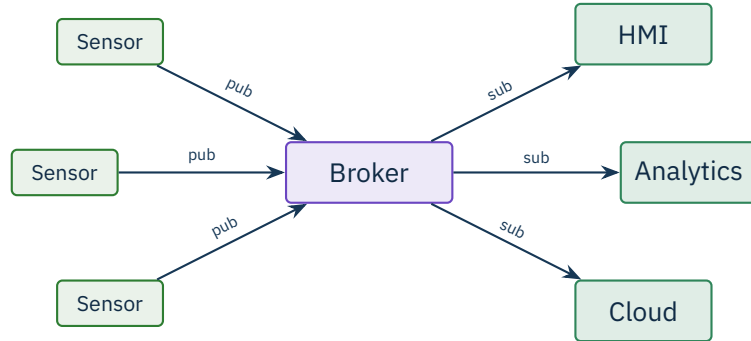
Source: IEC 62541 (multipart, OPC Foundation) | OPC UA Specification Part 2 (Security Model).

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└ Modern Secure Protocols  
  
└ OPC UA



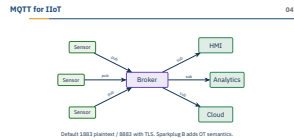
Sell OPC UA as the “protocol with seatbelts” — mutual X.509 authentication, signed messages, optional encryption, and a rich information model that lets you discover what a device *is*, not just what registers it has. Critical war story: the specification is over a dozen parts and a thousand pages; surveys (e.g. Kaspersky ICS-CERT, Claroty Team82 “OPC UA Deep Dive”) repeatedly find that a large share of real-world deployments run with security policy “None” and anonymous user tokens because that is the default in many configurators. Misconception: “OPC UA always uses TLS.” No — the binary transport has its own message-level crypto, and you must explicitly pick a SecurityPolicy like Basic256Sha256 or Aes256\_Sha256\_RsaPss; “None” is a valid choice. Class question: “how would you audit a plant to find anonymous-mode OPC UA servers?” Transition cue: “OPC UA is heavy — MQTT is the lighter modern option.”



Default 1883 plaintext / 8883 with TLS. Sparkplug B adds OT semantics.

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 └ Modern Secure Protocols  
 └ MQTT for IIoT



Frame MQTT as the publish/subscribe answer for IIoT: the broker is the choke point, sensors publish topics, anything authorised subscribes. Note the two ports – 1883 plaintext and 8883 with TLS – and remind students that Shodan finds hundreds of thousands of brokers on 1883 with no authentication and `#` wildcard subscriptions open. War story: in 2018 Avast scanned the open MQTT space and found brokers leaking smart-home, prison, and industrial telemetry; the situation has improved but not vanished. Sparkplug B (Eclipse Foundation) is worth mentioning because it adds the OT semantics MQTT lacks – state, birth/death certificates, structured payloads – and it is what most modern Unified Namespace deployments use. Misconception: “MQTT 5 is encrypted by default.” No – MQTT 5 added user properties and better error codes, not transport security; you still need TLS underneath. Transition cue: “let us put everything side by side.”



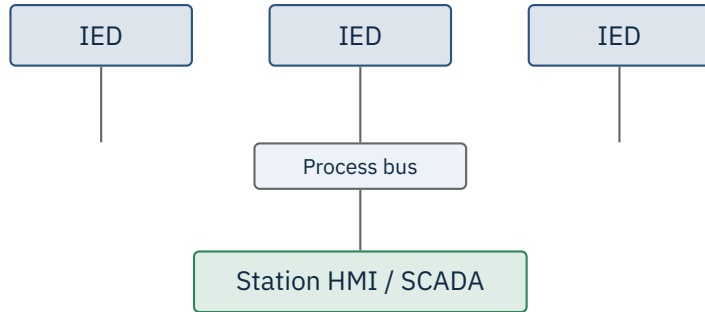
| Protocol    | Auth     | Integrity | Confid. | Notes               |
|-------------|----------|-----------|---------|---------------------|
| Modbus      | ×        | ×         | ×       | 1979, ubiquitous    |
| S7comm      | ×        | ×         | ×       | Siemens-proprietary |
| DNP3        | opt SAv5 | opt       | ×       | Utilities           |
| EtherNet/IP | opt CIP  | opt       | opt     | Rockwell            |
| Profinet    | weak     | ×         | ×       | Siemens / IRT       |
| OPC UA      | X.509    | sign      | TLS     | modern default      |
| MQTT        | +TLS     | TLS       | TLS     | IIoT messaging      |

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- Industrial Security (Bachelor)
  - Modern Secure Protocols
    - Protocol Security at a Glance

| Protocol    | Auth     | Integrity | Confid. | Notes               |
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| Modbus      | ×        | ×         | ×       | 1979, ubiquitous    |
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| OPC UA      | X.509    | sign      | TLS     | modern default      |
| MQTT        | +TLS     | TLS       | TLS     | IIoT messaging      |

Use the table to make a single argument: the security column is mostly a story about *defaults*. The protocols marked “opt” (DNP3 SAv5, CIP Security, MQTT+TLS) technically support strong crypto, but the boxes ship without it and the engineers configuring them rarely turn it on. Only OPC UA was designed with a non-trivial default — and even then operators routinely downgrade to SecurityPolicy “None” to fix interoperability. Misconception to puncture: “the green cells are safe protocols.” No, they are protocols where safety is possible; whether you got it depends on the deployment. Ask the class: “of these seven, which one would you put on a new greenfield project and why?” — there is no single right answer, the trade-off is determinism vs. security vs. ecosystem. Transition cue: “one application area pushes all of this to extremes — substations.”

**The substation stack**

**MMS** for client/server; **GOOSE** for sub-4 ms trip messages; **Sampled Values** for measurements. Goose has *no* authentication by default.

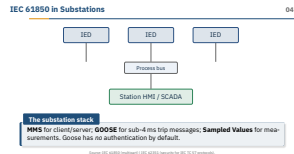
Source: IEC 61850 (multipart) | IEC 62351 (security for IEC TC 57 protocols).

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└ Modern Secure Protocols

└ IEC 61850 in Substations

Walk through the three services in order of latency budget: MMS for slow client/server, GOOSE for fast trip and interlock signals (layer 2 multicast, target end-to-end under 4 ms), and Sampled Values streaming raw current and voltage at the sampling rate of the merging unit. War story: a spoofed GOOSE frame can trip a breaker because the receiver only checks the state number and sequence; this exact attack family was demonstrated by Hoyos *et al.* and later operationalised by the CRASHOVERRIDE/Industroyer malware against the Ukrainian grid in 2016. IEC 62351-6 specifies authentication for GOOSE and SV, but few merging units actually implement it and switching it on adds latency the protection engineer may not tolerate. Misconception: “GOOSE is just multicast Ethernet, not a security concern.” It is exactly the lack of crypto on a high-trust safety bus that makes it interesting. Transition cue: “let us pull the section together.”



### ★ Key Takeaway: What to remember from this section

- Most legacy industrial protocols are unauthenticated and unencrypted by design.
- Modbus is the canonical example – 12 bytes of MBAP and a one-byte function code.
- Modern protocols (OPC UA, MQTT-TLS) bring real cryptography, but require correct configuration.
- Network-layer mitigations (firewalls, segmentation) compensate where the protocol cannot.

Land the section on one sentence: the protocol layer almost never gives you the security you want, so the compensating controls live in the network and in the process. Reinforce the Modbus number — 7-byte MBAP plus a one-byte function code is enough to read or write any register in any device on a bus, and that is the threat model most students will meet in their first OT engagement. Flag that the next section moves from *what is on the wire* to *how to put up walls around the wire*: firewalls, conduits, the Purdue model. If there is time, take one question on which protocol surprised them most — it is usually OPC UA, because the gap between specification and default deployment is the largest.

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- Modbus Organization. *Modbus Application Protocol Specification* v1.1b3, 2012.
- Modbus Organization. *Modbus/TCP Security*, 2018.
- IEEE 1815-2012. *Electric Power Systems Communications – DNP3*.
- IEC 60870-5-104:2006 + Amd. 1:2016. *Telecontrol equipment – Network access for IEC 60870-5-101*.
- IEC 62541 (multipart). *OPC Unified Architecture* (OPC Foundation).
- IEC 61850 (multipart). *Communication networks and systems for power utility automation*.
- IEC 62351 (multipart). *Security for TC 57 protocols*.
- OASIS. *MQTT Version 5.0*, 2019.
- ODVA. *The CIP Networks Library, Volume 1-8*, CIP Security spec.
- Siemens. *Step7 & S7 communication protocols* (S7comm, S7commPlus).
- Wightman, R. *Project Basecamp – ICS Vulnerability Catalog*, Digital Bond, 2012.

# End of Section 04

Industrial Protocols

Questions and discussion

